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Pauline Chassonnery

Curriculum Vitæ

Academic Background

Dec 2023	**PhD in Applied Mathematics**, *Paul Sabatier University*, Toulouse, France
2015 – 2020	**ENS Paris-Saclay Degree in mathematics, speciality International Research**, *École Normale Supérieure de Paris-Saclay*, France
July 2020	**Master's degree in Mathematics of Modelling, major Mathematics applied to Biological and Medical Sciences**, *Sorbonne University*, Paris, France
July 2018	**Master's degree in Mathematics Secondary-school and University Teaching**, *École Normale Supérieure de Cachan*, France
2012 – 2015	**CPGE Physics and Chemistry**, *Aristide Briand High School*, Évreux, France

Research Experiences

Postdoctoral fellowship (June 2024 – June 2026)

Title	**High performance simulation of Thermo-Hydro-Mechanical models in faulted geological systems.**
Project	ANR Earth–Beat.
Institutes	Bureau de Recherches Géologiques et Minières, Orléans & LJAD, Université Côte d'Azur, Nice.

PhD Thesis (September 2020 – December 2023)

Title	**Mathematical 3D modelling of connective tissues architecture emergence.**
Supervisors	Louis CASTEILLA and Diane PEURICHARD.
Institutes	Restore Institute, University Toulouse 3 & LJLL, Sorbonne University (France).

Master 2 intership (April – July 2020)

Title	**Mathematical 3D modelling of adipose tissue morphogenesis.**
Supervisor	Diane PEURICHARD.
Institute	Laboratoire Jacques-Louis Lions, Sorbonne University (France).

Predoctoral year of research abroad (October 2018 – June 2019)

Title	**Dynamical, multi-body interaction in a dense stellar system : formation of a super-massive black-hole.**
Supervisor	Roberto CAPUZZO-DOLCETTA.
Institute	Astrophysics Department, University of Roma La Sapienza (Italy).

Master 1 internship (April – July 2017)

Title	**Integrating primitive tumors's shape to cancer modelization by a level-set approach.**
Supervisors	Annabelle COLLIN and Olivier SAUT.
Institute	Inria team MONC & Bordeaux Institute of Mathematics (France).

Bachelor internship (spring 2016)

Title	**An interactive internet solver for the Riemann problem in digital fluid dynamic.**
Supervisors	Matthieu ANCELLIN and Jean-Michel GHIDAGLIA.
Institute	Center for MathematicaL studies and their Applications, ENS Cachan (France).

Teaching Experiences

2020 – 2022 **Numerical series and Function series**, *Bachelor 2 of Mathematics, Sorbonne University*, Paris, France.
Teaching assistant, 112 hours total in french, both remote (2020) and face-to-face (2021 and 2022).
- Tutorial sessions (12 × 3h per group, 1 group per year during 3 years)
- Creation of teaching material (methodological sheets, extensive answers to the exercices, example of numerical application in Python)
- Preparation of the test questions
- Grading of the tests and the mid-semester exam

2020 – 2022 **Numerical analysis**, *Bachelor 3 of Mathematics, Sorbonne University*, Paris, France.
Teaching assistant, 84 hours total in french, both remote (2020) and face-to-face (2021 and 2022).
- Practical sessions in Python (6 × 2h per group, 7 groups over 3 years)
- Creation of two sujects for the practical sessions out of six
- Grading of the practical exam

2025 **Analysis**, *common foundation of the Bachelor 1 of Sciences, Orléans University*, Orléans, France.
Teaching assistant, 20 hours in french.

2025 **Mathematics for Informatic**, *Bachelor 1 of Informatic, Orléans University*, Orléans, France.
Teaching assistant, 20 hours in french.

Collective responsabilities

2021 Organisation of the CARe graduate school workshop "Regeneration and Senescence : From Biological Mechanisms to Numerical Tools", *IUCT-Oncopole*, Toulouse, France.
- Choice of the theme
- Invitation of the external speakers and associated administrative procedures
- Design of the day's program
- Choice of the date, booking of the room and caterer
- Design and distribution of a promotional poster

Publications

[1] Pauline Chassonnery, Roberto Capuzzo-Dolcetta and Seppo Mikkola. "ARWV Code User Manual", 2019. arXiv:1910.05202.

[2] Pauline Chassonnery and Roberto Capuzzo-Dolcetta. "Dynamics of a superdense cluster of black holes and the formation of the Galactic supermassive black hole". *Monthly Notices of the Royal Astronomical Society*, 504(3):3909–3921, Apr 2021. doi:10.1093/mnras/stab1016.

[3] Pauline Chassonnery, Jenny Paupert, Anne Lorsignol, Childérick Sévérac, Marielle Ousset, Pierre Degond, Louis Casteilla and Diane Peurichard. "Fiber crosslinking drives the emergence of order in a three-dimensional dynamical network model". *Royal Society Open Science*, 11:231 456, Jan 2024. doi:10.1098/rsos.231456.

[4] Pauline Chassonnery, Jenny Paupert, Anne Lorsignol, Childérick Sévérac, Marielle Ousset, Louis Casteilla and Diane Peurichard. "Emergence and robustness of 3D complex tissue architectures", (submitted).

[5] Pauline Chassonnery, Diane Peurichard and Sinan Haliyo. "3D visualisation and segmentation tools for systems of spherical and rod-like objects", (in preparation).

[6] Nathan Bodereau, Théophile Lohier, Alban Moradell-Casellas, Damien Devismes, Lina Jolivet, Claire Aupart, Pauline Chassonnery, Florian Trichard, Nicolas Gilardi, Daniel Monfort-Climent and Anne-Marie Desaulty. "Analysing chemical maps of powders: application of machine learning in the framework of Lithium (Li) traceability", (in preparation).

Oral Communications and Posters

Feb. 2026 **Implementing contact-mechanics in faulted porous media using the Virtual Element Method.** Talk in french at the seminar of the Earth Sciences Institute of Orléans, 19th February 2026.

Nov. 2025 **Implementing contact-mechanics in faulted porous media using the Virtual Element Method.** Oral communication in french at the 17ème Journées d'Études des Milieux Poreux, 4th November 2025 in Orléans.

Sept. 2025 **Thermo-Hydro-Mechanical modelling of faulted geothermal reservoirs – Implementing solid mechanics in the framework of ComPASS.** Poster in english presented at the annual seminar of BRGM's scientific program GÉONUM, 6th September 2025 in Orléans.

Juin 2025 **Implementing contact-mechanics using VEM-bubbles in the ComPASS framework**. Oral communication in english at the workshop "Numerical Methods for Multi-Physics Problems" organized by the Porous Media Group of Bergen University's Department of Mathematics, 10th June 2025 in Bergen (Norway).

Jan. 2024 **Modélisation mathématique de la morphogénèse du tissu adipeux.** Flash talk in french at the joint CBI-IMT workshop "Modeling of life in Toulouse: from the atom to the animal", 26th January 2024 in Toulouse.

Jan. 2023 **Mathematical 3D modelling of adipose tissue morphogenesis.** Talk in english at the CARe graduate school workshop "Numerical and technical tools to better investigate metabolism, inflammation and cancer", 26th January 2023 at the IUCT-Oncopole in Toulouse.

Sept. 2022 **Mathematical 3D modelling of adipose tissue morphogenesis.** Talk in englih at the 3rd edition of the Mathematical Biology on the Mediterranean Conference, 6th September 2022 in Forth (Crete).

June 2022 **Mathematical 3D modelling of adipose tissue morphogenesis.** Poster in english presented at the Restore Institute Scientific day, 21st June 2022 in Toulouse.

Languages

French Native

English Fluent, TOEFL certification with a mark of 640/677

Computer Skills

Programming Advanced skills in Python, Fortran and *C++*. Proficient with Matlab.

Proficient with the binding library nanobind the expose C++ types in Python.

Proficient with git versionning tools (bash interface, gitlab, continuous integration)

Writing Proficient with LaTeXand Beamer.

Proficient with the reference management software Zotero